

BITS PILANI
Work Integrated Learning Programme
M.Tech – Software Engineering

Introduction to DevOps (SEZG514)

Assignment 2

ACEest Fitness & Gym Management System

DevOps CI/CD Implementation Project

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Semester: 3

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Project Repository

GitHub: <https://github.com/2024TM93636/aceest-fitness-ci-cd>

DockerHub: <https://hub.docker.com/r/sarath93636/aceest-gym>

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1. Introduction

This project demonstrates the implementation of a complete DevOps CI/CD pipeline for the ACEest Fitness & Gym Management System. The objective is to automate the software development lifecycle to achieve faster delivery, improved code quality, and reliable deployments.

The application is developed using a Flask-based REST API and integrated with modern DevOps tools including GitHub for version control, GitHub Actions and Jenkins for CI/CD, Docker for containerization, Kubernetes (Minikube) for deployment, Pytest for testing, and SonarQube for code quality analysis.

This implementation simulates a real-world DevOps environment with automated workflows and advanced deployment strategies.

2. CI/CD Architecture Overview and Pipeline Stages

The CI/CD pipeline consists of the following stages:

1. Code Development

- Flask-based application
- Modular structure with:
 - Members management
 - Workouts
 - Search functionality

2. Version Control (GitHub)

Repository:

<https://github.com/2024TM93636/aceest-fitness-ci-cd>

Practices:

- Structured commits
- Version tagging (v1.0, v1.1, v2.0)
- Branching (main, assignment2)

3. Continuous Integration

GitHub Actions

- Triggered on every push
- Steps:
 - Install dependencies
 - Run Pytest
 - Build Docker image

Jenkins

- Pulls latest code
- Executes:
 - Build
 - Test
 - Docker image creation
- Uses Jenkinsfile for pipeline definition

4. Testing (Pytest)

- Automated unit tests for:
 - CRUD operations
 - Workouts
 - API validation
- Ensures early bug detection

5. Containerization (Docker)

Docker Images:

- sarath93636/aceest-gym:v1.0
- sarath93636/aceest-gym:v1.1
- sarath93636/aceest-gym:v2.0

Benefits:

- Environment consistency
- Easy deployment
- Version control

6. Deployment (Kubernetes – Minikube)

- Local cluster using Minikube
- YAML configurations:
 - Deployment
 - Service (NodePort)

Access command:

```
minikube service aceest-service
```

7. Deployment Strategies

Rolling Update

- Gradual pod update
- No downtime

Blue-Green Deployment

- Two environments
- Traffic switching

Canary Deployment

- Partial traffic to new version
- Risk minimized

A/B Testing

- Different versions for users
- Performance comparison

Shadow Deployment

- Runs in parallel
- No user impact

8. Code Quality (SonarQube)

- Static code analysis
- Metrics:
 - Bugs
 - Vulnerabilities
 - Code smells

- Coverage

Scanner executed using Docker:

```
docker run --rm -e  
SONAR_HOST_URL="http://host.docker.internal:9000" -e  
SONAR_TOKEN="your_token" -v "${PWD}:/usr/src" sonarsource/sonar-  
scanner-cli
```

3. Challenges Faced & Solutions

1. Docker Engine Issue

- Error: dockerDesktopLinuxEngine not found
- Solution: Restarted Docker and verified using:

```
docker run hello-world
```

2. Kubernetes Service Not Accessible

- Issue: No running pods
- Solution:

```
kubectl get pods  
kubectl describe pod <pod-name>
```

3. Git Detached HEAD

- Solution:

```
git branch assignment2 <commit-id>  
git checkout assignment2  
git push origin assignment2
```

4. SonarQube Authentication Error

- Solution: Used SONAR_TOKEN instead of SONAR_LOGIN

5. PowerShell Command Issues

- Solution: Used single-line Docker commands

4. Key Automation Outcomes

- Fully automated CI/CD pipeline
- Automated testing before deployment
- Containerized deployment
- Multi-version deployment
- Advanced deployment strategies implemented
- Code quality enforcement

5. Results & Benefits

- Faster deployment
- Reduced manual effort
- Improved code quality
- High reliability
- Zero-downtime deployments

6. Conclusion

This project successfully demonstrates a complete DevOps pipeline integrating CI/CD, testing, containerization, and Kubernetes deployment. The system ensures scalability, automation, and reliability, aligning with modern software engineering practices.

7. References

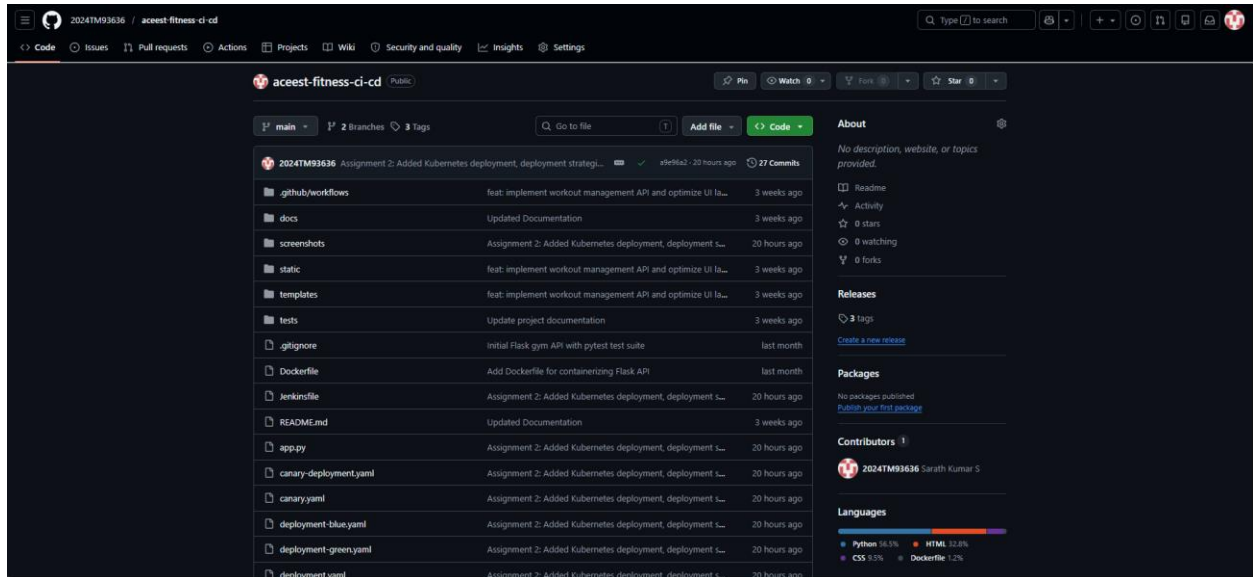
- Docker Documentation – <https://docs.docker.com>
- Kubernetes Documentation – <https://kubernetes.io/docs/>
- Jenkins Documentation – <https://www.jenkins.io/doc/>
- SonarQube Documentation – <https://docs.sonarsource.com/>
- GitHub Actions Documentation – <https://docs.github.com/actions>

Deployment Note:

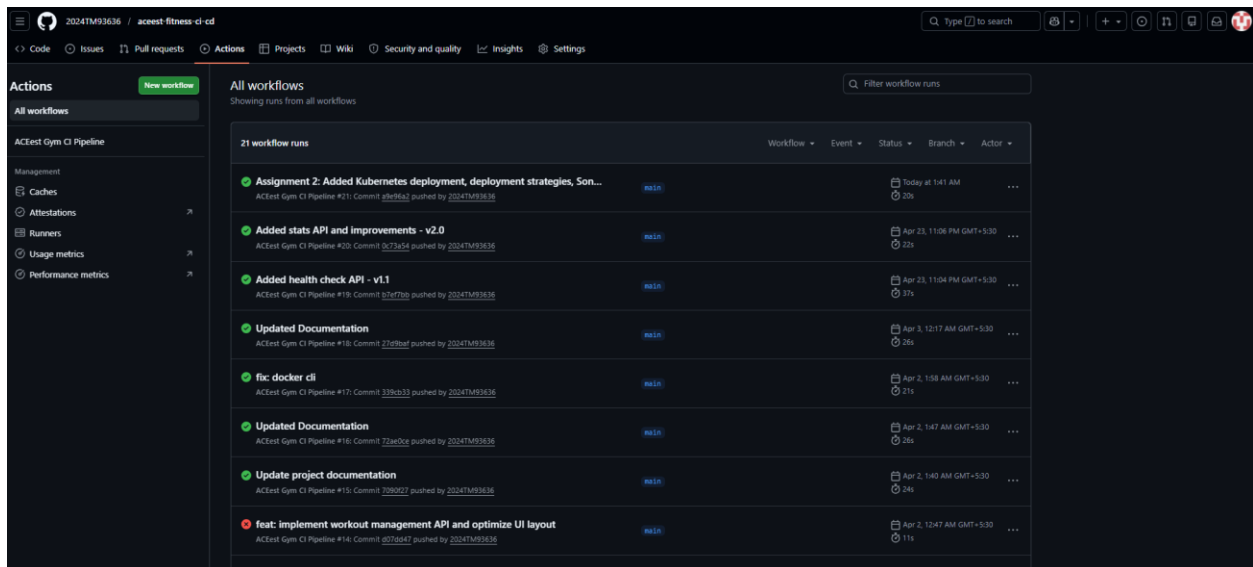
The application is deployed in a local Kubernetes environment using Minikube. Hence, the endpoint URL is accessible only within the local system. Screenshots and execution evidence are provided for validation.

8. Screenshots

1. GitHub Repository



2. GitHub Actions Pipeline



3. Jenkins Pipeline

The screenshot shows the Jenkins web interface for a pipeline named 'accest-fitness-build'. The build is #14, completed on April 1, 2026, at 8:44:02 PM. The console output shows the build was started by user 'admin' and completed successfully. The run spent 1 min 27 sec total from scheduled to completion, with 1 min 27 sec build duration. The build was triggered by a git push to the repository 'https://github.com/20247169363/accest-fitness-ci-ed.git'. The build steps include 'git checkout' and 'mvn clean install -DskipTests'. The build artifacts are available for download. The Jenkins version is 2.541.2.

Jenkins / accest-fitness-build / #14

#14 (Apr 1, 2026, 8:44:02 PM)

Started by user **admin**

This run spent:

- 1 min 27 sec waiting
- 1 min 27 sec build duration
- 1 min 27 sec total from scheduled to completion

git Revision: 339cb31430524c0b6e2eb29678458c1b460f735
Repository: <https://github.com/20247169363/accest-fitness-ci-ed.git>
• refs/remotes/origin/main

No changes.

REST API Jenkins 2.541.2

4. Docker Images

The screenshot shows the Docker Desktop interface. The left sidebar contains navigation options: Containers, Images, Volumes, Kubernetes, Builds, Models, MCP Toolkit, Docker Hub, Docker Scout, and Extensions. The main panel displays the 'Containers' section, showing a table of running containers. The table includes columns for Name, Container ID, Image, Port(s), CPU (%), Last started, and Actions. Three containers are listed: minikube, sonarqube, and upbeat_mirzakhani. The minikube container is using 64.75% CPU and 55.16MB memory. The sonarqube container is using 0% CPU and 0B memory. The upbeat_mirzakhani container is using 0% CPU and 0B memory.

docker.desktop PERSONAL

Containers [Give feedback](#)

Container CPU usage: 64.75% / 000% (0 CPUs available)

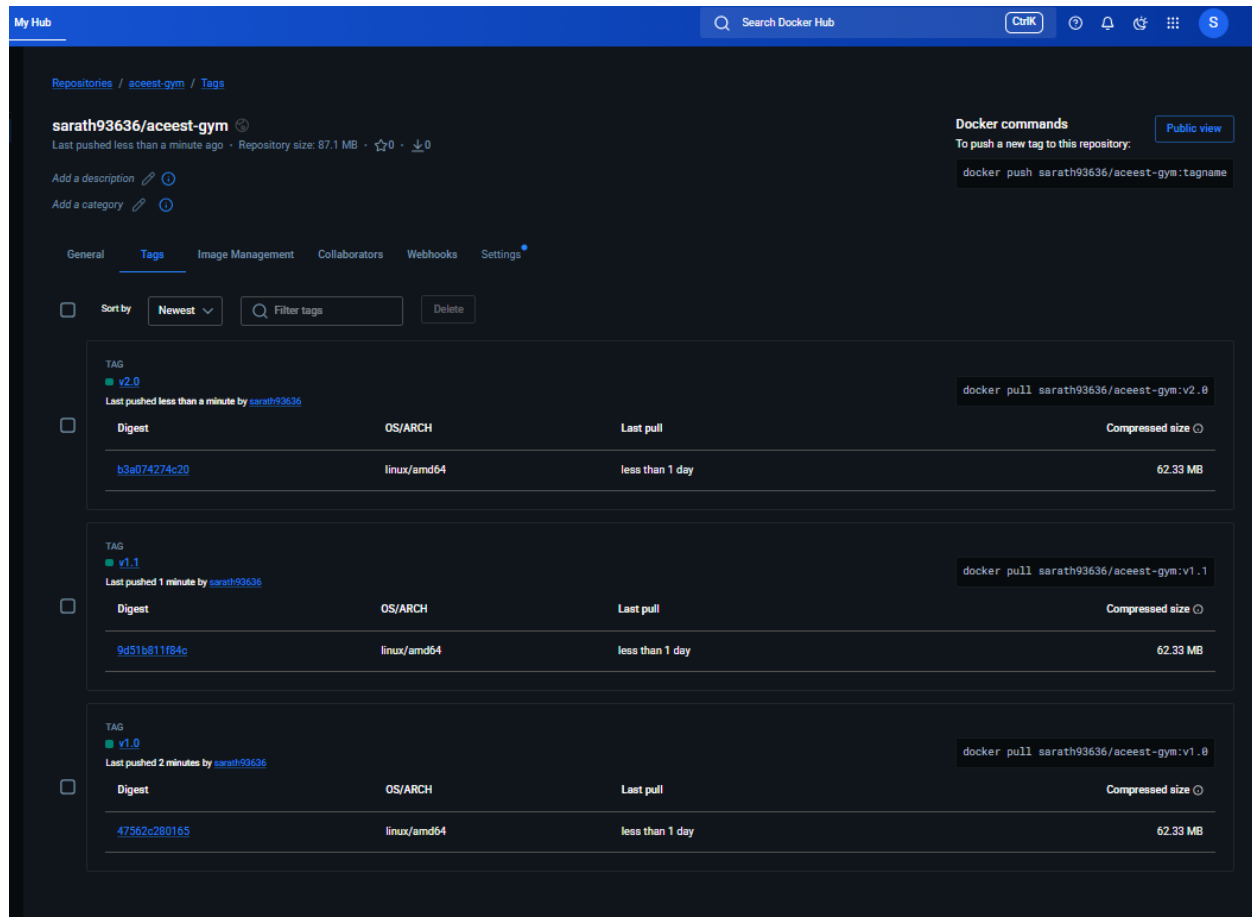
Container memory usage: 55.16MB / 0B

Search

Only show running containers

	Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐	minikube	7b52951e547c	k8s.minikube/kicbasev0.0.48	60577:22 Show all ports (5)	64.75%	2 hours ago	Refresh Stop Restart Delete
☐	sonarqube	9dba74b2d80e	sonarqube	9000:9000 Show all ports (5)	0%	59 minutes ago	Refresh Stop Restart Delete
☐	upbeat_mirzakhani	db1eb4b15a34	sonarsource/sonar-scanner-cli		0%	8 minutes ago	Refresh Stop Restart Delete

5. Docker Hub



The screenshot shows the Docker Hub interface for the repository `sarath93636/aceest-gym`. The page includes a search bar, repository details, and a list of tags. The tags section shows three tags: `v2.0`, `v1.1`, and `v1.0`. Each tag entry includes a table with columns for Digest, OS/ARCH, Last pull, and Compressed size. The `v2.0` tag has a digest of `b3a074274c20` and a size of 62.33 MB. The `v1.1` tag has a digest of `9d51b811f84e` and a size of 62.33 MB. The `v1.0` tag has a digest of `47562c280165` and a size of 62.33 MB. The page also includes a 'Docker commands' section with a 'Public view' button and a 'docker push' command.

Repositories / `aceest-gym` / Tags

sarath93636/aceest-gym ⓘ
Last pushed less than a minute ago · Repository size: 87.1 MB · ☆0 · ↓0

Add a description ⓘ ⓘ
Add a category ⓘ ⓘ

General **Tags** Image Management Collaborators Webhooks Settings ⓘ

☐ Sort by **Newest** ▾

Tag
v2.0
Last pushed less than a minute by sarath93636

`docker pull sarath93636/aceest-gym:v2.0`

Digest	OS/ARCH	Last pull	Compressed size ⓘ
b3a074274c20	linux/amd64	less than 1 day	62.33 MB

Tag
v1.1
Last pushed 1 minute by sarath93636

`docker pull sarath93636/aceest-gym:v1.1`

Digest	OS/ARCH	Last pull	Compressed size ⓘ
9d51b811f84e	linux/amd64	less than 1 day	62.33 MB

Tag
v1.0
Last pushed 2 minutes by sarath93636

`docker pull sarath93636/aceest-gym:v1.0`

Digest	OS/ARCH	Last pull	Compressed size ⓘ
47562c280165	linux/amd64	less than 1 day	62.33 MB

6. Kubernetes Pods

```
• (venv) PS D:\M.Tech\BITS Pilani\Semester 3\Introduction to Devops\Assignment\aceest-fitness-ci-cd> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
aceest-deployment-64f84576df-hblvz 1/1     Running   0           4m56s
aceest-green-55f79bc9bc-p7nnh       1/1     Running   0           3m51s
```

7. Kubernetes Services

```
• (venv) PS D:\M.Tech\BITS Pilani\Semester 3\Introduction to Devops\Assignment\aceest-fitness-ci-cd> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
aceest-deployment-64f84576df-tcc6f 1/1     Running   0           9m12s
• (venv) PS D:\M.Tech\BITS Pilani\Semester 3\Introduction to Devops\Assignment\aceest-fitness-ci-cd> kubectl get services
NAME          TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
aceest-service NodePort    10.97.97.154 <none>        80:30007/TCP    10m
kubernetes    ClusterIP   10.96.0.1    <none>        443/TCP          12m
```

8. Running Application

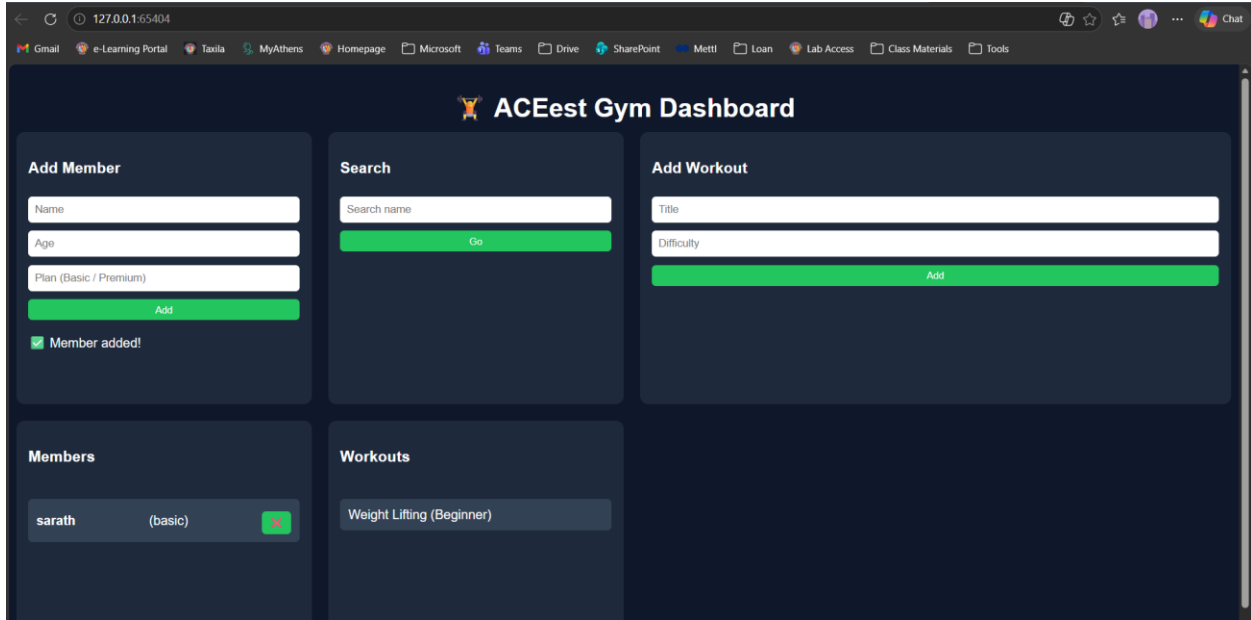
```
PS D:\M.Tech\BITS Pilani\Semester 3\Introduction to Devops\Assignment\aceest-fitness-ci-cd> minikube service aceest-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	aceest-service	80	http://192.168.49.2:30007

Starting tunnel for service aceest-service.

NAMESPACE	NAME	TARGET PORT	URL
default	aceest-service		http://127.0.0.1:65404

Starting tunnel for service aceest-service.
Opening service default/aceest-service in default browser...
Because you are using a Docker driver on windows, the terminal needs to be open to run it.



9. Deployment Strategy Proof

```
(venv) PS D:\M.Tech\BITS Pilani\Semester 3\Introduction to Devops\Assignment\aceest-fitness-ci-cd> kubectl set image deployment/aceest-deployment aceest-container=sarath93636/aceest-gym:v1.1
deployment.apps/aceest-deployment image updated
(venv) PS D:\M.Tech\BITS Pilani\Semester 3\Introduction to Devops\Assignment\aceest-fitness-ci-cd> kubectl rollout status deployment/aceest-deployment
deployment "aceest-deployment" successfully rolled out
(venv) PS D:\M.Tech\BITS Pilani\Semester 3\Introduction to Devops\Assignment\aceest-fitness-ci-cd> kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
aceest-deployment-7c9bb5d7bd-m7kr5	1/1	Running	0	30s

Rolling Update - New pod created and old pod terminated without downtime

```
(venv) PS D:\M.Tech\BITS Pilani\Semester 3\Introduction to Devops\Assignment\aceest-fitness-ci-cd> kubectl rollout undo deployment/aceest-deployment
deployment.apps/aceest-deployment rolled back
```

Rollback - Successfully reverted to previous stable version

```
(venv) PS D:\M.Tech\BITS Pilani\Semester 3\Introduction to Devops\Assignment\aceest-fitness-ci-cd> kubectl scale deployment aceest-deployment --replicas=3
deployment.apps/aceest-deployment scaled
(venv) PS D:\M.Tech\BITS Pilani\Semester 3\Introduction to Devops\Assignment\aceest-fitness-ci-cd> kubectl apply -f canary.yaml
deployment.apps/aceest-canary created
(venv) PS D:\M.Tech\BITS Pilani\Semester 3\Introduction to Devops\Assignment\aceest-fitness-ci-cd> kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
aceest-canary-75674c8fb6-vj7ws	1/1	Running	0	11s
aceest-deployment-64f84576df-5bcbx	1/1	Running	0	39s
aceest-deployment-64f84576df-hblvz	1/1	Running	0	6m52s
aceest-deployment-64f84576df-vktbg	1/1	Running	0	39s
aceest-green-55f79bc9bc-p7nrh	1/1	Running	0	5m47s

Canary Deployment - Partial traffic routed to new version

```

• (venv) PS D:\M.Tech\BITS Pilani\Semester 3\Introduction to Devops\Assignment\aceest-fitness-ci-cd> kubectl apply -f deployment-blue.yaml
deployment.apps/aceest-blue created
• (venv) PS D:\M.Tech\BITS Pilani\Semester 3\Introduction to Devops\Assignment\aceest-fitness-ci-cd> kubectl apply -f deployment-green.yaml
deployment.apps/aceest-green unchanged
• (venv) PS D:\M.Tech\BITS Pilani\Semester 3\Introduction to Devops\Assignment\aceest-fitness-ci-cd> kubectl get pods --show-labels
NAME                                READY   STATUS    RESTARTS   AGE   LABELS
aceest-blue-6bb866f4bf-jbr6n        1/1     Running   0           57s   app=aceest,pod-template-hash=6bb866f4bf,version=blue
aceest-canary-75674c8fb6-vj7ws       1/1     Running   0          3m56s   app=aceest,pod-template-hash=75674c8fb6,version=canary
aceest-deployment-64f84576df-5bcbx   1/1     Running   0          4m24s   app=aceest,pod-template-hash=64f84576df
aceest-deployment-64f84576df-hblvz   1/1     Running   0          10m     app=aceest,pod-template-hash=64f84576df
aceest-deployment-64f84576df-vktbg   1/1     Running   0          4m24s   app=aceest,pod-template-hash=64f84576df
aceest-green-55f79bc9bc-p7nnh        1/1     Running   0          9m32s   app=aceest,pod-template-hash=55f79bc9bc,version=green
• (venv) PS D:\M.Tech\BITS Pilani\Semester 3\Introduction to Devops\Assignment\aceest-fitness-ci-cd> kubectl edit service aceest-service
Edit cancelled, no changes made.

```

Blue-Green Deployment - Traffic switched between environments

```

• (venv) PS D:\M.Tech\BITS Pilani\Semester 3\Introduction to Devops\Assignment\aceest-fitness-ci-cd> kubectl apply -f v1-deployment.yaml
deployment.apps/aceest-v1 created
• (venv) PS D:\M.Tech\BITS Pilani\Semester 3\Introduction to Devops\Assignment\aceest-fitness-ci-cd> kubectl apply -f v2-deployment.yaml
deployment.apps/aceest-v2 created
• (venv) PS D:\M.Tech\BITS Pilani\Semester 3\Introduction to Devops\Assignment\aceest-fitness-ci-cd> kubectl apply -f service-v1.yaml
service/aceest-service-v1 created
• (venv) PS D:\M.Tech\BITS Pilani\Semester 3\Introduction to Devops\Assignment\aceest-fitness-ci-cd> kubectl apply -f service-v2.yaml
service/aceest-service-v2 created
• (venv) PS D:\M.Tech\BITS Pilani\Semester 3\Introduction to Devops\Assignment\aceest-fitness-ci-cd> minikube service aceest-service-v1
>> minikube service aceest-service-v2

```

NAMESPACE	NAME	TARGET PORT	URL
default	aceest-service-v1	80	http://192.168.49.2:32458

Starting tunnel for service aceest-service-v1.

NAMESPACE	NAME	TARGET PORT	URL
default	aceest-service-v1		http://127.0.0.1:60546

Starting tunnel for service aceest-service-v1.

Opening service default/aceest-service-v1 in default browser...

! Because you are using a Docker driver on windows, the terminal needs to be open to run it.

A/B Testing - Different versions served to different users

```

• (venv) PS D:\M.Tech\BITS Pilani\Semester 3\Introduction to Devops\Assignment\aceest-fitness-ci-cd> kubectl port-forward deployment/aceest-v2 5001:5000
Forwarding from 127.0.0.1:5001 -> 5000
Forwarding from [::1]:5001 -> 5000
Handling connection for 5001
Handling connection for 5001
Handling connection for 5001
Handling connection for 5001
Handling connection for 5001

```

Shadow Deployment - New version running in parallel without affecting users

10. SonarQube Dashboard

sonar-project.properties

1 sonar.projectKey=aceest-gym
2 sonar.projectName=ACEest Gym
3 sonar.projectVersion=1.0
4
5 sonar.sources=.
6 sonar.exclusions=venv/**,tests/**,__pycache__/**
7
8 sonar.python.version=3.10

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SONARQUBE

(venv) PS D:\M.Tech\BITS Pilani\Semester 3\Introduction to Devops\Assignment\aceest-fitness-ci-cd> docker run --rm -e SONAR_HOST_URL="http://host.docker.internal:9000" -e SONAR_TOKEN="sq_621914f" 1f8db66ee34f3267008d6ce9b6f57c83 v "4(nao):/usr/src" sonarsource/sonar-scanner-cli
19:50:05.285 INFO Scanner configuration file: /opt/sonar-scanner/conf/sonar-scanner.properties
19:50:05.294 INFO Project root configuration file: /usr/src/sonar-project.properties
19:50:05.383 INFO SonarScanner CLI 8.0.1.6346
19:50:05.397 INFO Linux 6.6.87.2-microsoft-standard-WSL2 amd64
19:50:09.222 INFO Communicating with SonarQube Community Build 26.4.0.121862
19:50:09.223 INFO JRE provisioning: os[linux], arch[x86_64]
19:50:34.827 INFO Starting SonarScanner Engine...
19:50:34.832 INFO Java 21.0.9 Eclipse Adoptium (64-bit)
19:50:38.720 INFO Load global settings
19:50:39.053 INFO Load global settings (done) | time=334ms
19:50:39.060 INFO Server id: 1478411E-A227tzqhwEyjwLX8LU
19:50:39.077 INFO Loading required plugins
19:50:39.078 INFO Load plugins index
19:50:39.211 INFO Load plugins index (done) | time=133ms
19:50:39.212 INFO Load/download plugins
19:50:43.433 INFO Load/download plugins (done) | time=4178ms
19:50:44.211 INFO Process project properties
19:50:44.254 INFO Process project properties (done) | time=44ms
19:50:44.285 INFO Project key: aceest-gym
19:50:44.286 INFO Base dir: /usr/src
19:50:44.287 INFO Working dir: /tmp/.scannerwork
19:50:44.316 INFO Load project settings for component key: 'aceest-gym'
19:50:44.430 INFO Load quality profiles
19:50:46.303 INFO Load quality profiles (done) | time=1873ms
19:50:46.446 INFO Load active rules

sonarqube

Projects Issues Rules Quality Profiles Quality Gates Administration

Search for projects... A

aceest-gym ☆ Passed

Last analysis: April 24, 2026, 1:24 AM | Version 2.0

main

Overview Issues Security Hotspots Measures Code Activity

Project Settings Project Information

Quality Gate

Passed

All conditions passed.

Measures

Bugs 0 A Reliability

Vulnerabilities 0 A Security

Security Hotspots 0 A Security Review

Code Smells 3 A Maintainability

Coverage 92.1% on 1.1k lines to cover

Duplications 1.2% on 1.6k lines

Lines of Code 2.1k Python, YAML, Docker, CSS, HTML

Issues

0 A Blocker

0 A Critical

0 A Major

3 A Minor

0 A Info

New Issues since last analysis
0 Added

Quality Gate (Default Sonar way) Passed

Bugs is less than or equal to 0

Vulnerabilities is less than or equal to 0

Hotspots Reviewed is greater than or equal to 80.0%

3 conditions

About This Project

ACEest Fitness & Gym Management System

Python 5 Languages

2.1k Lines of Code

Project Key
aceest-gym

Details

Project Activity

100

50

0

Apr 17 Apr 18 Apr 19 Apr 20 Apr 24

More activity